

Design Summary and Recommendations

This document presents the current design choices and architectural decisions for building a scalable repository of K–12 educational images. The objective is to identify an efficient system for collecting, storing, and retrieving image metadata while maintaining low latency and reliable performance for applications that will use this dataset.

First, the team evaluates potential **image repositories** that provide educational and openly licensed images. After reviewing several sources, **Wikimedia Commons** is selected as the primary repository because it contains a large collection of freely licensed educational images and provides strong API support for programmatic access. The **Wikimedia API** allows the system to ingest images and metadata based on categories, topics, or search keywords relevant to K–12 learning.

Next, the system design focuses on an efficient **database architecture** for storing image metadata. Instead of storing full image files in the database, the system stores **metadata such as image title, description, source URL, category, license information, and thumbnail links**, while the actual images remain hosted on Wikimedia's content delivery network. This approach reduces storage requirements and improves scalability. **MySQL** is chosen as the primary database because it offers reliable performance, strong indexing capabilities, and seamless integration with Python-based ingestion scripts.

To support efficient retrieval of images, the system implements **indexed database tables and optimized SQL queries** that allow images to be searched by keywords, topics, and categories. This indexing structure improves query performance and enables the system to handle larger datasets as the repository grows.

Another key focus is **latency analysis**. The system identifies potential latency sources, including API response times from Wikimedia, database query performance, and image loading delays when retrieving images from remote repositories. Understanding these factors helps guide design decisions such as indexing, query optimization, and possible caching strategies to ensure faster retrieval of images.

Based on this evaluation, the recommended architecture includes the following components:

1. **Wikimedia Commons as the primary image source**, providing openly licensed educational images.
2. **Python-based ingestion scripts** to collect and process image metadata automatically through the Wikimedia API.
3. **MySQL database** to store structured image metadata with indexed fields for efficient retrieval.

4. **External hosting of images on Wikimedia servers**, storing only metadata and URLs locally to reduce storage and infrastructure costs.
5. **Performance optimization strategies**, including indexed queries and potential caching, to minimize latency and improve user experience.

This design supports a **scalable, efficient, and maintainable system** that enables quick retrieval of educational images and can be extended in the future to support advanced search, topic classification, or integration with AI-powered educational tools.

Relevant Links

Wikimedia Commons: <https://commons.wikimedia.org>

Wikimedia API Documentation: https://www.mediawiki.org/wiki/API:Main_page

MySQL Documentation: <https://dev.mysql.com/doc/>

Project Repository: https://github.com/hhaider3/SER517_team5_industry